



UTKARSH

Scientist / Engineer 'SC', Cryogenic Propulsion · Liquid Propulsion Systems Centre, ISRO

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Research Statement

Propulsion scientist at ISRO's Liquid Propulsion Systems Centre, working on cryogenic engine development for the Indian space program. My work spans **cryogenic system transients, thermal chill-down, and the fabrication technologies that make cryogenic hardware possible** — engine cycle and power-balance analysis, start and shutdown transient simulation, pressure-surge modelling, subsystem realisation through additive manufacturing and vacuum brazing, and the critical analysis of test anomalies that turn them into design fixes. I am seeking a doctoral position at CERN to apply this experience of cryogenic fluids, transients and advanced joining to accelerator-scale systems. I have been a long-time admirer of cutting-edge work carried out at CERN and would be grateful to join as a doctoral student.

Research & Professional Experience

Aug 2023

– Present

Scientist / Engineer 'SC' — Cryogenic Propulsion

Liquid Propulsion Systems Centre, Indian Space Research Organization ·
Thiruvananthapuram, India

- Engine cycle analysis, optimisation and power-balance simulations
- Evaluation of thermal chill-down performance
- Engine start and shutdown sequence transient simulations
- Subsystem-level development through additive manufacturing and brazing
- Flight performance prediction and data analysis
- Critical analysis of test anomalies and development of prevention mechanisms

Feb 2023

– Jul 2023

Executive Trainee (Engineer)

NTPC Limited (Maharatna PSU) · India

- Trained in operation and maintenance of fluid and thermal systems of an 800 MW supercritical thermal power plant.
- Completed a three-week certification at the National Power Training Institute, Ministry of Power, Faridabad.

Jan 2022

– May 2022

Product Design Intern

Indian Railways · India

- Designed an ergonomic exercise bike for chair cars and waiting rooms, dimensioned from Indian anthropometric percentile data.

May 2021
– Jun 2021

Research Intern

Indian Institute of Technology Indore · under Dr. Anand, Dean, IIT Indore

- Structural and dynamic study of tennis racket design — frame stiffness, vibration response and material selection.

Education

2018 – 2022

B.Tech., Production Engineering

Birla Institute of Technology, Mesra · CGPA 8.59 / 10 — First Class with Distinction

2017

Senior Secondary (CBSE)

Shivam Public School, Nalanda · 86.8%

2015

Secondary (CBSE)

Jawahar Navodaya Vidyalaya, Nalanda · 10.0 CGPA

Publications

Dec 2025

Utkarsh, Arif, M., Santhosh Kumar, Lyjith, K. "Experimental and Computational Study of Copper–Silver Filler Brazed Joint for Aerospace Applications." *Conference paper*.

Dec 2025

Utkarsh, Nithin, P., Lyjith, K. "Modelling of Rocket Engine Thrust Chamber Brazing Dynamics." *Conference paper*.

Apr 2024

Utkarsh, Dhanunjaya, P., Nithin, P., Lyjith, K., Arumugam, M. "Investigation of Cryogenic Engine Thrust Chamber Forming." *Conference paper*.

Accepted
(Nov 2025)

Utkarsh, Adarsh, A. N., Varrier, M., Bijukumar, K. S. "Analytical Modelling of Entrapped Gas Driven Diaphragm Rupture Signature in Start Fuel Ampoule of a Rocket Engine." *24th ISME Conference on Mechanical Engineering — Innovations for Viksit Bharat (ISME 2026)*.

Accepted
(Nov 2025)

Utkarsh, Adarsh, A. N., Varrier, M., Bijukumar, K. S. "Numerical Modelling of Pressure Surges during Rapid Pressurisation of Cryogenic Rocket Engine Measurement Canalisation Line." *30th National Conference on Cryogenics and Superconductivity*.

Accepted
(Dec 2025)

Utkarsh, Adarsh, A. N., Varrier, M., Bijukumar, K. S. "Experimental and Numerical Study of Gas–Gas Interaction of Air–Butane Combusted Flow." *30th National Conference on Cryogenics and Superconductivity*.

Patents Applied

Lead PI	A Methodology to Obtain a Sound CuCrZrTi to 12–10PH Dissimilar Braze Joint Lead Principal Investigator · Application filed, ISRO / Liquid Propulsion Systems Centre (reviewed and cleared at center level)
Lead PI	Design of Single Exit Clustered Thrust Chamber for Rocket Application Lead Principal Investigator · Application filed, ISRO / Liquid Propulsion Systems Centre (reviewed and cleared at center level)

Technical Skills

DOMAINS	Cryogenic propulsion · Vacuum brazing & dissimilar joining · System engineering · Aerospace materials · Structural & thermal FEA · Ergonomics
ADVANCED	SolidWorks, Mechanical Design
INTERMEDIATE	ANSYS Workbench & Fluent, Additive Manufacturing
BASIC	Python, MATLAB, C, CATIA V5, Abaqus

Honours & Awards

2022, 2023	Graduate Aptitude Test in Engineering (Mechanical) — All India Rank 119 (2023); All India Rank 211 (2022), among ~100,000 candidates.
2020	Runner-up, Institute Innovation Contest , BIT Mesra — ergonomic ground chair for street vegetable vendors, ranked second best innovation of the university.
2020	Runner-up, Campus Innovathon — national case-study competition; podium finish among 2,000 teams for sustainable fuel-retail station design.

Selected Project & Leadership

2019 – 2021	GARUD — Electric Formula Student Racecar, Team Srijan, BIT Mesra Design Executive — Chassis & Ergonomics - Designed the spaceframe chassis, cockpit, headrest, ergonomic seat, steering wheel, motor test rig and welding fixture for BIT Mesra's first electric FSAE car. - Ranked 10th overall and 10th in engineering design at Formula Bharat 2021 — the only first-year car in the top ten.
2020 – 2022	Co-founder & Joint President, Society for Industrial Management and Engineering BIT Mesra — founded and led a 100+ member technical society for industrial engineering practice.

Certifications

Dassault	SolidWorks Mechanical Design Associate — Dassault Systèmes, Germany
AMCAT	Data Processing Specialist & Business Analyst — Aspiring Minds